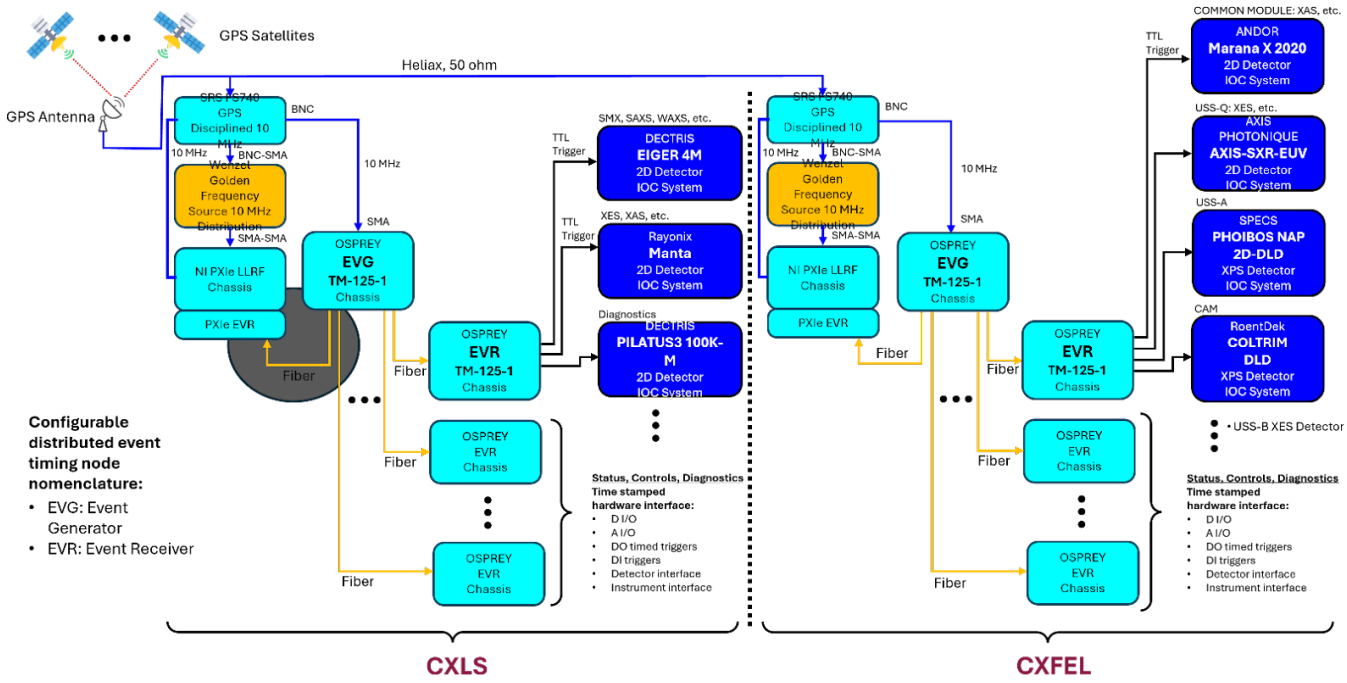
1. ASU Timing Network



The ASU timing network distributes triggers and timestamps to all subsystems, enabling precise coordination of system actions and accurate correlation of data across the control infrastructure. Core components include a GPS-synchronized reference clock, a 10 MHz RF master oscillator, an FPGA-based Event Generator (EVG) to issue deterministic triggers and time-of-day events, and distributed Event Receivers (EVRs) that deliver synchronized triggers and timestamps to instruments and data acquisition systems.

2. GPS Receiver

The GPS receiver provides a Network Time Protocol (NTP) server for distributing absolute time to control system hosts and Event Receivers (EVRs). It also outputs a 1 Pulse Per Second (1PPS) digital signal, which serves as a precise hardware reference for the start of each second. This pulse is used by the Event Generator to align distributed timestamps across the facility.

3. Event Generator (EVG)

The Event Generator is an FPGA-based timing master responsible for generating and distributing synchronized event codes and timing metadata across the facility. It accepts the following inputs:

- A 1 Hz timing pulse from the GPS receiver (1PPS)
- Up to two RF signals for phase alignment with the primary 10 MHz RF master oscillator
- Up to eleven fiber optic outputs for distributing the encoded event stream to Event Receivers (EVRs)

The EVG aligns event emission with the RF clock to ensure deterministic, low-jitter trigger distribution. It serves as the central point for coordinating timing across beamline, laser, LLRF, and detector subsystems.

Specifications – from Eric – jitter, resolution, RF master oscillator, drift.

Specifications for the EVR – rise time, number of signals, signal types, pulse width limitations.

Notes from Mark Holl:

The EVG may be the master timer. Currently, there is a DG645
The DG645 has less than 25 picoseconds of jitter, 1 MHz trigger rates 8 delay outputs
The EVG would set the beam path
The EVG would send triggers at the beam rate to: LLRF, Modulators, LLRF gates, Laser to create beam, High Power Laser, and experiments.

4. Event Receivers (EVRs)

Event Receivers (EVRs) listen to the timing signal sent from the Event Generator (EVG). This signal contains a continuous stream of 16-bit words, where the first 8 bits identify an event code (such as “start acquisition”), and the second 8 bits include related data or metadata.

These event codes are predefined and used to trigger actions across the facility. The data portion of the stream may carry status values or control information. When the EVG sends a full message or set of values, it ends with a special “message complete” event code so the EVR knows full data is ready to use.

One of the most important pieces of information sent this way is the Time-of-Day, which tells each EVR what the current second is, based on a common “EPOCH” starting point. A 1Hz pulse updates this once per second, and each EVR keeps track of the time between those pulses using its precise internal clock. This gives each system the ability to timestamp data with nanosecond-level accuracy and align it with other systems. The timing stream can also include useful status information like beam current, machine protection state, source and destination beamlines, and other metadata that helps interpret the collected data later.

EVRs come in two forms. Some are embedded directly into devices, such as digitizers, where they provide internal triggers and attach timestamps directly to each measurement. Others are standalone units that output digital pulses (typically TTL signals) to trigger external devices like detectors or lasers that need precise timing but don’t generate their timestamps.

This system ensures all parts of the facility stay in sync even when data is coming from many independent sources.

5. Integration and timing with various DAQ equipment

The timing system is integrated into both the accelerator control infrastructure and each experimental environment to ensure all subsystems operate in sync. In typical installations, the Event Generator (EVG) is located near the GPS receiver and reference clock, while the Event Receivers (EVRs) are positioned close to the instruments or data acquisition systems they control. This arrangement minimizes latency and ensures tight coordination between triggers and timestamped data. This document focuses on how timing is integrated with several key DAQ systems at ASU, including the Quartz chassis, the PXI-based LLRF system, a ROENTek detector, and a Dectris Eiger detector. Each case highlights how triggers and timestamps are delivered to ensure synchronized data acquisition.

6. Timing and DAQ Using the Quartz Chassis

The Quartz chassis is a modular DAQ system developed for use in accelerator diagnostics. Each unit includes an EVG, an EVR, fast protection I/O, and a 32-channel, 24-bit ADC. In typical applications, the EVR provides synchronized triggers, and all 32 ADC channels are timestamped in alignment with the facility timing system.

6.1. Quartz Chassis – DAQ-32-250-24



Key Features

- 32 differential channel 24-bit analog inputs
- Sample rates up to 250kS/s
- Real time monitoring data streaming at 20Hz
- AC/DC coupling
- Synchronized events across multiple chassis via timing fiber
- Ethernet control and data monitoring
- Rack mountable (2U form factor)
- Open source project
- EPICS support

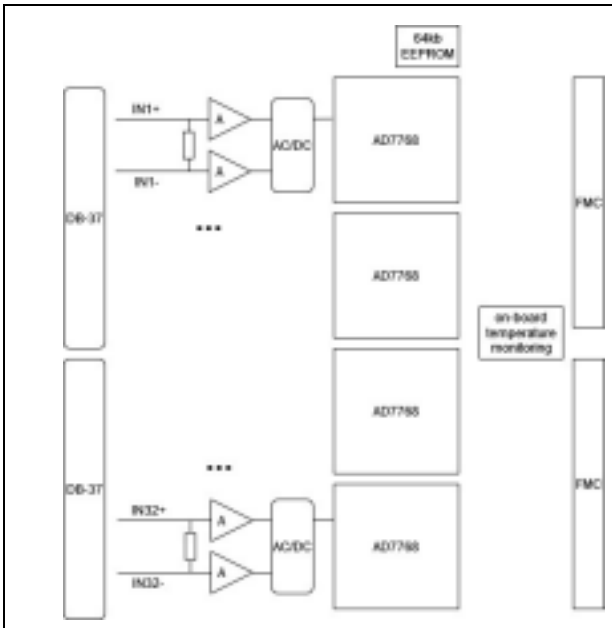
The Osprey DAQ-32-250-24 is a 32 channel 250 kHz SR24 bit resolution analog voltage digitizing device packaged in a 2U standard rack mount chassis. It is the IO front end for a full DAQ system that are synchronized and time stamped using a GPS receiver that synchronizes the all chassis through a fiber timing network. A full integration through the EPICS control system provides real time monitoring of all configuration and internal status to ensure reliable operation. Systems are operational that acquire data at configurable frequencies from 1 kHz to 250 kHz and transmit and store data up to 1,024 signals at 250,000 samples per second.

The chassis houses an FMC based on the Marble FMC developed by Berkeley National Laboratory (https://github.com/BerkeleyLab/Marble) and an Osprey Quartz ADC board with 32 channels of 24-bit ADCs. Analog inputs are provided on the front panel using two DB37 connectors.

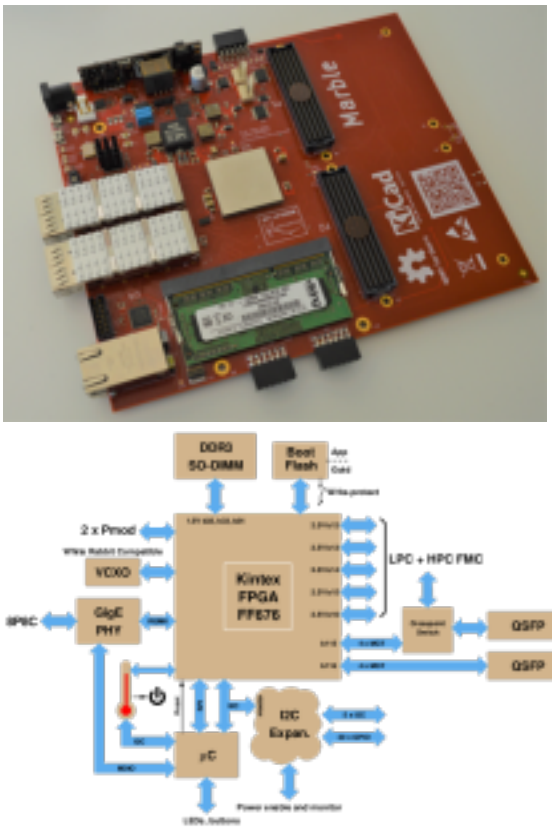
Designed for ease of maintenance and serviceability, the chassis is held together via 18 countersunk screws and also provides fiber and ethernet connections for chassis to chassis and server communication. With an on-board,

user-programmable FPGA, the DAQ is designed to function as an front-end field instrument with on-board signal processing and application communications node in a third-party distributed data or control control system application (such as an EPICS).

Basic Architecture



Osprey Quartz ADC Board Overview



Marble Board Overview

The Osprey Quartz ADC board contains four Analog Devices AD7768 ADCs of which the rest of the board is built around. Inputs at the DB-37 connectors are scaled and pass through filtering and a per-channel selectable DC blocking stage before being sent to an ADC. The ADC then streams its data via LVDS through the FMC connectors to the Marble Board.

The Marble board is built around a Kintex 160T 7 Series FPGA. This is one of the highest performing FPGAs available from AMD while allowing use of the license-free tool chain. The Marble board is responsible for monitoring, packaging, and reporting all streamed data from the ADCs.

The front panel is a 3.48 x 19 x 0.128 rack mountable panel equipped with 2 LC fiber connectors, an LED power light and 2 handles. There are 2 cutouts that expose the 2 DB-37 connectors attached to the Quartz ADC 250kHz board. The front panel attaches to the chassis via 6 countersunk screws.



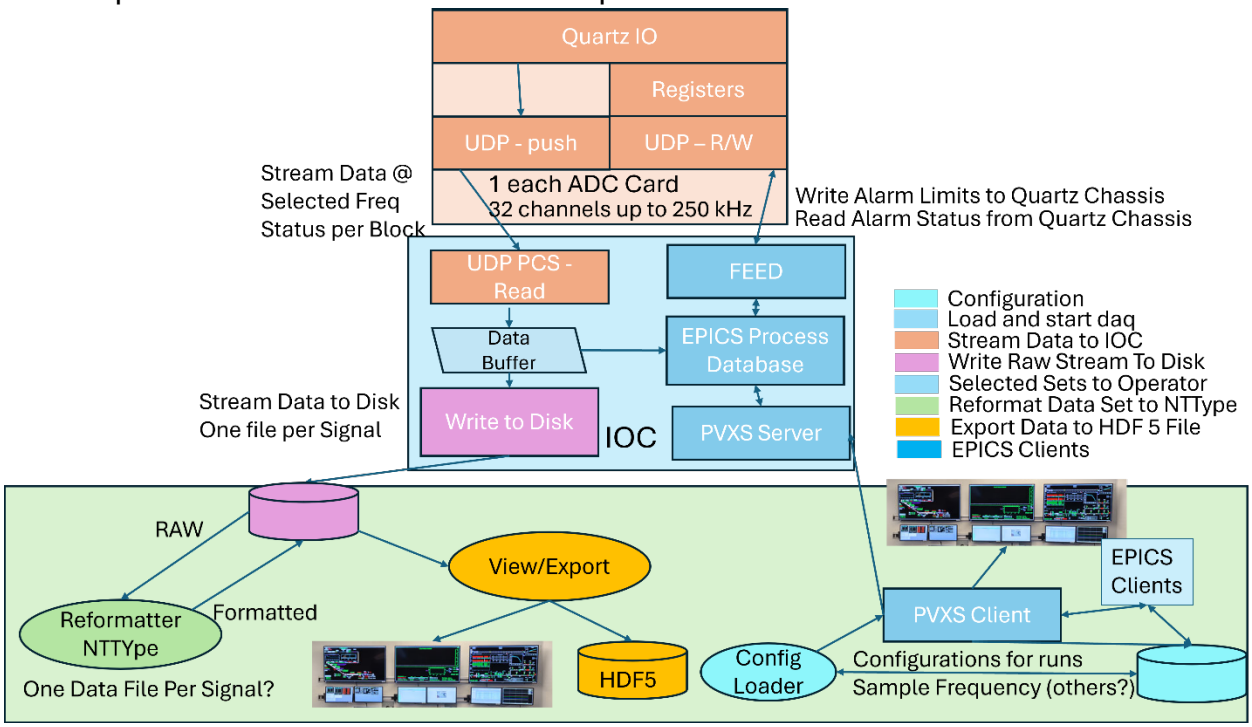
The back panel is 16.8x3x0.128 equipped with a DC power connector, a power switch, a 5A fuse, an Ethernet port, and a fan that is pulling the air from inside the chassis. Note: there is a fan on the side of the chassis that is blowing air directly across the heatsink on the FPGA board. The back panel attaches to the chassis via 4 countersunk screws



6.2 The Architecture for the quartz chassis

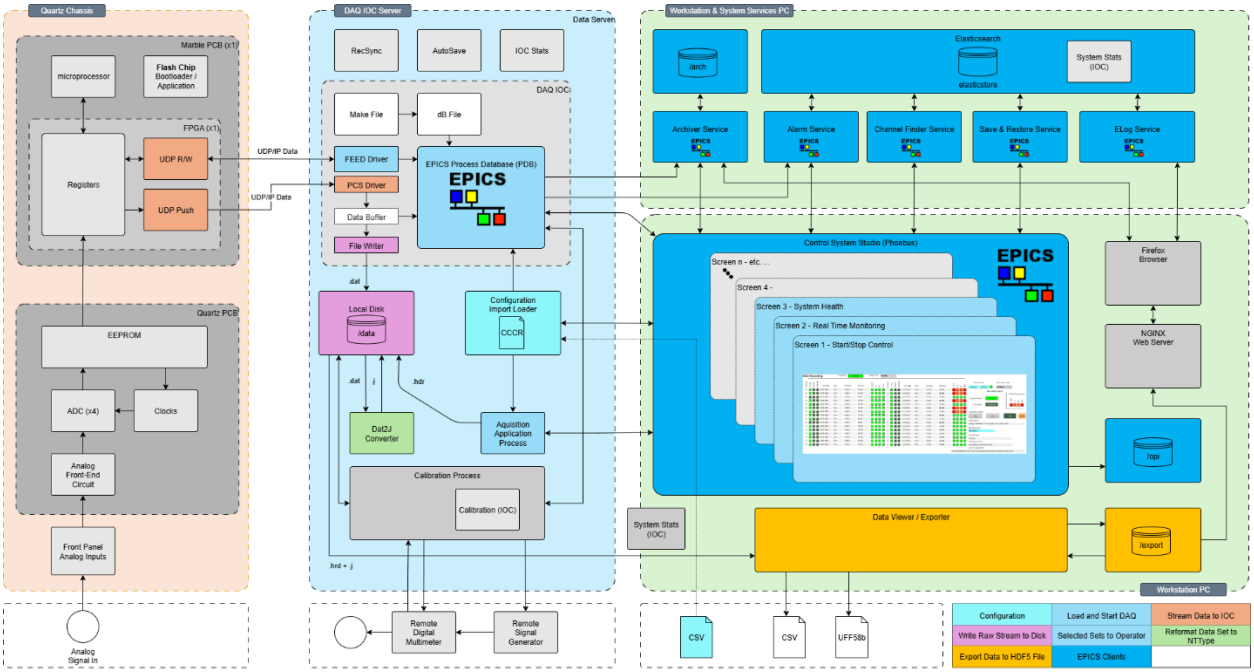
The EVR is embedded in the Quartz chassis. A fiber into the front panel bring the events and data into the FPGA environment. Time stamps are derived from the data stream. Acquisition triggers are derived from the Event stream. These links can be daisy-chained to other Quartz chassis. The fiber connectors are on the front panel.

There are 32 channels of -10V to +10V signals. They also plug into the front panel with 16 channels on each connector.
There are 4 digital inputs and 4 digital outputs on the rear panel. The digital inputs can be used to integrate TTL faults into the fast protection chain implemented in the quartz chassis. The 4 digital outputs can be used to set each of 4 different fault levels that can be used to put equipment into a safe state,.
The software architecture is shown in the picture below. There are many aspects to the data acquisition that are color coded to help track the data flow. .



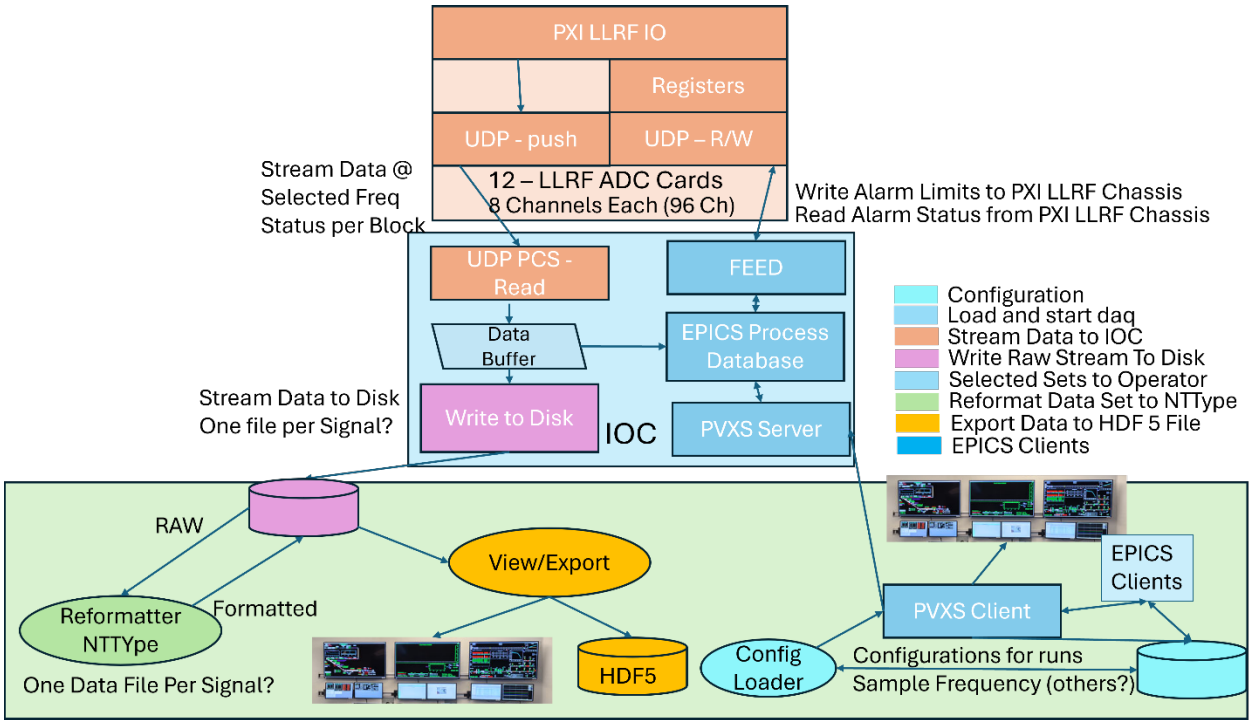
In this second drawing, the software is more detailed. The software is housed in three different pieces of equipment: an FPGA based Quartz chassis on the left backlit in peach, a I/O Control Server in the middle in light blue and a server computer in light green on the right. Each software function is also color coded including: loading a configuration into the Quartz Chassis, Loading and starting the DAQ using a standard Phoebus screen in EPICS. Once the DAQ is enabled, the data is streamed off of the Quartz Chassis over a dedicated serial link into an data

buffer on the IOC server. All selected data is written to a disk on the IOC server. In addition, the data buffer feeds an EPICS Database to present selected data at a selected rate to the operator so that the data can be monitored in real time through a PHOEBUS display (a standard EPICS client). Once data collection is complete, the compressed data (24 bits per sample) are placed into an NTType with even byte alignment for the data and saved to disk. The reformatted data can be viewed, selected and exported into an HDF5 file. Throughout this process, there are approximately 1,024 signals on the Quartz chassis that can be monitored and archived with standard EPICS clients that represent the state of health of the hardware.



6.3 LLRF System

The timing for the LLRF system is derived directly to the PXI hardware that controls the LLRF. In addition to the internal time, there is an EVR module that is plugged into the PXI crate that provides time stamp and synchronization events to the LLRF chassis. The data acquisition is expected to perform similarly to the Quartz chassis. The streaming of the data off of the PXI create must be demonstrated. The storage and monitoring is not yet defined.



6.4 Dectris Detector (Eiger)

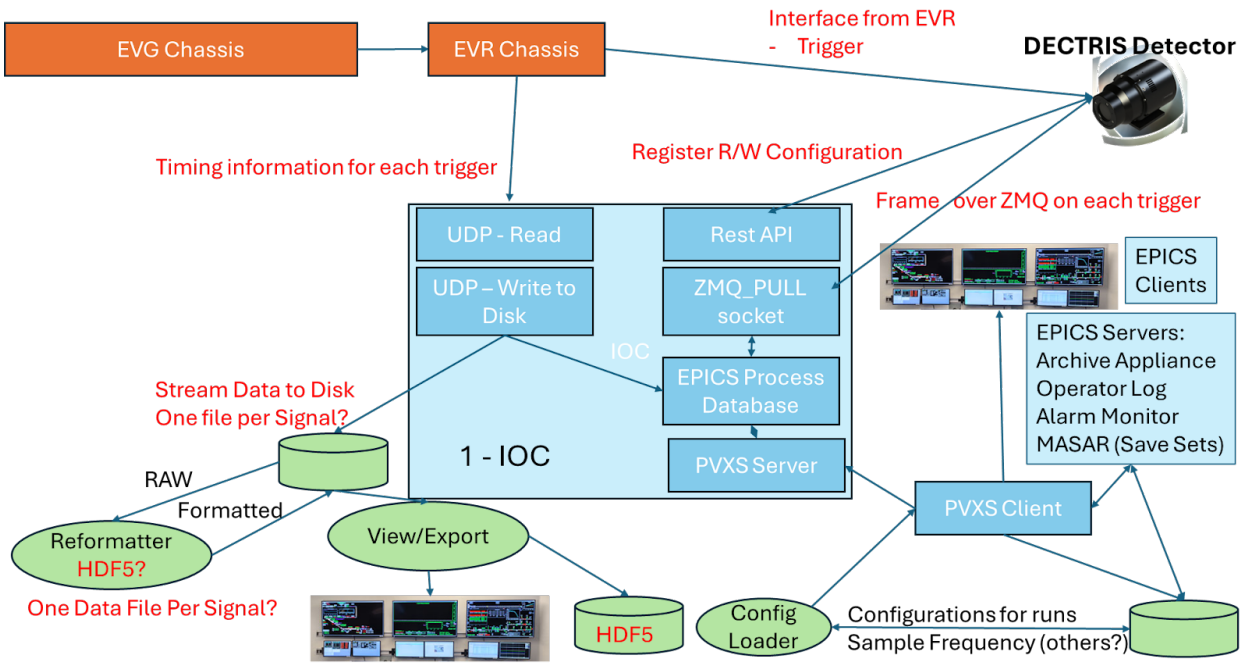
The Dectris Eiger detector requires external hardware triggers to acquire frames. These triggers are typically delivered via TTL signals from an EVR, often housed in a Marble-based chassis. Alternatively, in test setups or less time-critical scenarios, a standalone timing unit such as a DG645 may be used for trigger generation.

In hybrid mode, the detector operates using both its internal FileWriter and the ZMQ-based streaming interface (Stream2). Each trigger initiates a frame acquisition, and the detector assigns a unique internal timestamp or frame ID to each image. However, these timestamps are not synchronized to the facility's global timebase.

To address this, the IOC receives real-time frame data over the ZMQ Stream2 interface and uses a custom plugin (under development) to inject timestamps derived from the EVR event stream. This ensures that each frame is aligned with the global timing system and can be correlated with other systems in the experiment.

The FileWriter on the detector simultaneously stores raw image slices locally as a backup. These files can optionally be pulled by the IOC after acquisition and reprocessed to apply EVR-based timestamps in post.

This architecture ensures low-latency data access for monitoring and analysis, while preserving a redundant copy of all data for reprocessing and validation.

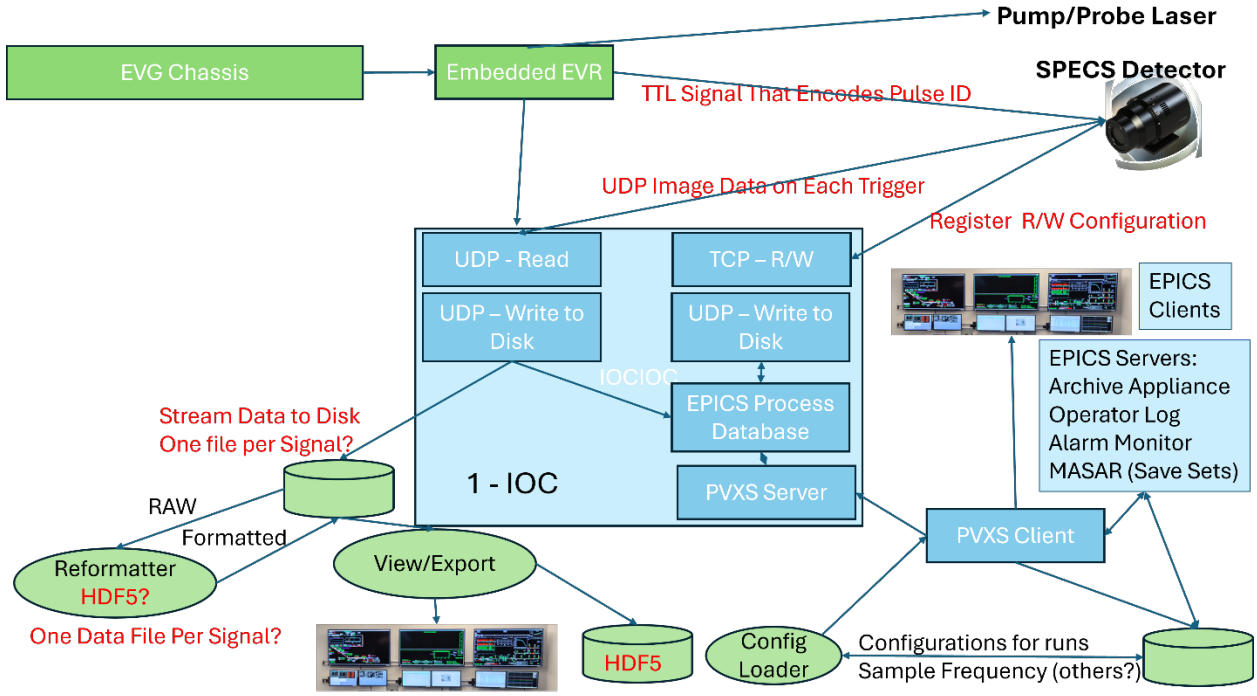


Several technical details are still under evaluation, including:

- How precise the Eiger's internal timestamps are, and whether they exhibit frame drift or dropped frames under high-rate acquisition
- The final format and structure of the data packets received over ZMQ
- Whether timestamping and data merging are performed inline in the IOC or buffered with a fixed delay
- The required buffer size and memory footprint to support streaming rates up to 500 Hz, with ~4 MB per frame and a peak sustained bandwidth of 20 GBps

6.4 RoenTek/SPECS Detectors

These detector require external triggers. The triggers provided to the detector may come from an EVR housed in a Marble Board Chassis, with fast digital outputs to provide TTL type signals to the Dectris Device. Alternatively, a DG645 may be used.' There is also a digital signal that will be wired to an available ToF signal that changes at 1kHz and is used as part of the data from the detector to identify the subsecond portion of the timestamp to support data correlation.



Notes:
1 SPECS Detector Detector
1 table of ToF data?
1 kHz trigger rate to initiate image acquisition
Pump/Probe Laser Triggers
1 ttl signal that places a user ID onto the data array in one column
Size?
characterize time to take and xfer

Data Acquisition Buffers
Table – 2D or 3D
SPECS channel - Time Stamp Accuracy? Precision? Missing Frames?
EVR must provide a stream of events with time stamp to the IOC
What is the structure of the data packet?
Are the time stamp and data stream merged in the IOC during ACQ

Configuration and Status Registers
TCP/IP communication to the PXI Controller
What is the protocol for accessing the registers

For the SPECS controller

- Configuration parameters
- Control parameters
- Monitor parameters